Multiple Features and Isolation Forest-Based Fast Anomaly Detector for Hyperspectral Imagery

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***Abstract*— Hyperspectral anomaly detection (HAD) has drawn a significant attention of late due to its importance in many military and civilian applications. In this article, a fast hyperspectral anomaly detector that combines multiple features and isolation forest is proposed. This approach, which is based on the assumption that the anomalous pixels are more susceptible to isolation than the background pixels, consists of two main parts. First, the spectral, Gabor, extended morphological profile (EMP) and extended multiattribute profile (EMAP) features are extracted from the hyperspectral image (HSI). Next, the isolation forest of each feature is constructed using the subsampling strategy. This combination of multiple features can exploit both the spectral and spatial information of the HSI, thereby improving the anomaly detection performance significantly. Compared with eight state-of-the-art HAD methods, the experimental results on four real hyperspectral data sets demonstrate that the performance of our proposed approach is quite competitive in terms of detection accuracy and running time.**

***Index Terms*— Hyperspectral anomaly detection (HAD), hyperspectral image (HSI), isolation forest, multiple features.**

I. Introduction

BY COMBINING spectroscopic technology with imaging technology, hyperspectral remote sensing can obtain the spatial and spectral continuous hyperspectral images (HSIs) [1]–[3]. Compared with conventional optical or multispectral images, HSIs provide rich spectral and spatial information for use in monitoring the Earth's surface and achieving fine identification of various land cover materials. HSIs have thus seen a widespread use in many remote sensing applications, including classification [3], unmixing [4], clustering [5], change detection [6], and target or anomaly detection [2], [7].

Among these applications, hyperspectral anomaly detection (HAD) has drawn a significant attention due to its importance in many military and civilian applications, such as military surveillance, mineral exploration, environmental monitoring, and so on. Compared with the background pixels, anomalous pixels have two remarkable characteristics: their low probability of occurrence and a spectral signature that differs from that of the background pixels. HAD can be modeled as an unsupervised binary classification task distinguishing the anomaly class from the background class. It aims to label the anomalous pixels or background pixels without any prior knowledge.

Over the past three decades, researchers have proposed a number of HAD algorithms. Among them, the well-known Reed–Xiaoli (RX) algorithm [8], proposed by Reed and Xiaoli, has become the benchmark for anomaly detection in HSIs. The RX algorithm assumes that the background pixels obey a multivariate Gaussian distribution and has two versions: the global RX (GRX), which models the background using the whole image, and the local RX (LRX) which models the background using the local dual-window image. However, the Gaussian distribution assumption is inadequate for modeling the intricate background in real-world HSIs. Several variants of the RX algorithm have been presented [9]–[17]. The Kernel RX (KRX) [9] algorithm is a nonlinear version of the RX and characterizes the nonGaussian distributions in high-dimensional feature space by using the kernel theory. The cluster-based anomaly detection (CBAD) algorithm [10] partitions the whole image into several clusters and then detects anomalies in each cluster by using the RX algorithm. Moreover, the more recently proposed weighted RX (WRX) [13] models the background by assigning a specific weight to each pixel. In [17], the cluster KRX (CKRX) algorithm was proposed to speed up the KRX algorithm. The main idea behind the CKRX algorithm involves grouping the whole image into several clusters and then applying a fast eigenvalue decomposition algorithm to generate the anomaly detection index.

In addition to the RX algorithm and its modified versions, a number of representation-based algorithms have also

Manuscript received October 22, 2019; revised January 17, 2020 and February 14, 2020; accepted March 2, 2020. Date of publication March 18, 2020; date of current version August 28, 2020. This work was supported in part by the National Key Research and Development Program of China under Grant 2018YFB1403501, in part by the National Natural Science Foundation of China under Grant 61936014, Grant 61772427, Grant 61751202 and Grant U1803263, in part by the Key Area Research and Development Program of Shaanxi Province under Grant 2019ZDLGY17-07, and in part by the Fundamental Research Funds for the Central Universities under Grant G2019KY0501 and Grant 3102019PJ006. *(Corresponding author: Feiping Nie.)*

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Digital Object Identifier 10.1109/TGRS.2020.2978491

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drawn substantial attention; these include the collaborative representation-based algorithms [18]–[22], the sparse representation-based algorithms [23]–[29], and the low-rank representation-based algorithms [30]–[36]. Li and Du [18] proposed a collaborative representation-based anomaly detection (CRD) algorithm for the HSIs, which assumes that each background pixel can be approximately represented by its surrounding pixels in a local dual window, whereas this is not the case for anomalous pixels. To further benefit from the spatial information of the HSI, a morphology-based collaborative representation detector (MCRD) was proposed in [19]. Moreover, a CRD algorithm with principal component analysis for removing outliers (PCAroCRD) [21] was proposed to reduce the errors of judgment when the test pixel is an anomalous pixel, and several pixels from its surrounding pixels are similarly anomalous pixels. Recently, the recursive CRD (RCRD) algorithm [22] was proposed to reduce the computational complexity of the CRD algorithm; here, the recursive update is derived by a matrix inversion lemma. Chen *et al.* [23] proposed a sparse representation-based anomaly detection algorithm that operates on the assumption that each background pixel can be represented by a few atoms in the dictionary. The background joint sparse representation (BJSR) [24] algorithm first selects the most representative background pixels using the joint sparsity model and then adopts sparse representation for the test pixel based on the selected background pixels. Zhao *et al.* [26] proposed a novel sparsity score estimation framework for HAD. Li *et al.* [27] presented a novel HAD algorithm capable of exploiting the structured sparsity in order to model the background more accurately. The low-rank and sparse representation (LRASR) algorithm, proposed in [33], exploits the low-rank property of the background pixels to distinguish sparse anomalies. Sun *et al.* [31] proposed the low-rank and sparse matrix decomposition (LRaSMD) algorithm, which scores each pixel according to the Euclidean distance between the corresponding sparse component vector and the mean vector of the sparse matrix. The low-rank and sparse matrix decomposition-based Mahalanobis distance (LSMAD) algorithm explores the low-rank prior knowledge of the background pixels and scores each pixel according to the Mahalanobis distance. Moreover, HAD algorithms based on support vector data description (SVDD) [37], [38], morphological and attribute filters [39]–[41], tensor decomposition [42]–[46], deep convolutional neural networks [47]–[50], and so on have also been explored, indicating that the research in this field is highly active.

In a departure from the previous articles, a novel multiple features and isolation forest-based detector (MFIFD) for hyperspectral imagery is proposed in this article. In recent years, Gabor feature [51], extended morphological profile (EMP) feature [52], and extended multiattribute profile (EMAP) feature [53], [54] have been successfully applied in the hyperspectral classification task. The Gabor feature can effectively represent the spatial texture information of HSI. The EMP and EMAP features can effectively represent the spatial structure information of HSI. First, the spectral, Gabor, EMP, and EMAP features of the HSI are extracted. Any single feature only contains a specific spectral or spatial characteristic of the HSI; accordingly, the combination of multiple spectral and spatial features offers significant potential improvements for the HAD process. Subsequently, the isolation forest for each feature is built using the subsampling strategy. The isolation forest was presented by Liu *et al.* [55], [56] 10 years ago and first introduced into the HAD field by Li *et al.* [57]. The rationale behind its use is that the anomalous pixels are more susceptible to isolation than the background pixels. It is worthwhile to highlight the main contributions of this article, which are as follows.

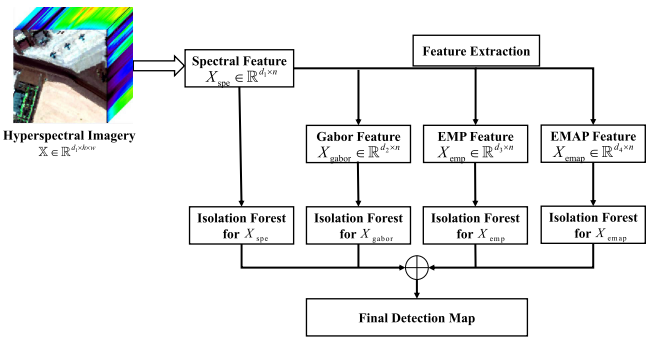


Fig. 1. Flowchart of the proposed MFIFD algorithm.

1) The spectral, Gabor, EMP, and EMAP features contain the specific spectral and spatial characteristics of the HSI. The complementary properties of different spectral and spatial features are combined with the isolation forest. It is demonstrated that the combination of multiple spectral and spatial features could significantly improve the anomaly detection performance.
2) Extensive experiments on four real data sets demonstrate that the performance of the proposed MFIFD algorithm is quite competitive in terms of detection accuracy and running time.

The remainder of this article is organized as follows. Section II briefly reviews the extraction of multiple spectral and spatial features from the HSI and then describes the proposed MFIFD algorithm in detail. In Section III, we conduct extensive experiments on four real data sets to demonstrate the effectiveness and superiority of the proposed MFIFD algorithm. Finally, we conclude this article in Section IV.

II. Multiple Feature and Isolation Tree-Based HAD

Let $\mathbb{X} \in \mathbb{R}^{d_1 \times h \times w}$ denotes the HSI data, where d_1 is the number of spectral bands, whereas h and w are the height and width of the image scene, respectively. For the sake of convenience, the 3-D HSI data $\mathbb{X}$ is transformed into a 2-D matrix $\boldsymbol{X}_{\text{spe}} = [\boldsymbol{x}^1_{\text{spe}}, \boldsymbol{x}^2_{\text{spe}}, \ldots, \boldsymbol{x}^n_{\text{spe}}] \in \mathbb{R}^{d_1 \times n}$, where each column of $\boldsymbol{X}_{\text{spe}}$ is a spectral pixel vector in the HSI, and $n = h \times w$ is the number of pixels. Fig. 1 shows the overall flowchart of the proposed algorithm.

A. Multiple Feature Extraction

For each pixel, we extract four types of commonly used features in hyperspectral imagery processing: the spectral feature,

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the Gabor feature, the EMP feature, and the EMAP feature. More details of these four features are given as follows:

1) *Spectral Feature:* The spectral feature of each pixel is its spectral signature, and $\boldsymbol{X}_{\text{spe}} \in \mathbb{R}^{d_1 \times n}$ denotes the spectral feature matrix.
2) *Gabor Feature [51]:* To extract the Gabor feature of each pixel, we first obtain the projection matrix $\boldsymbol{W} \in^{d_1 \times m}$ by solving the standard principal component analysis (PCA) model

$$\max_{\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}} \operatorname{tr}\left(\boldsymbol{W}^T \boldsymbol{X}_{\text{spe}} \boldsymbol{X}_{\text{spe}}^T \boldsymbol{W}\right) \tag{1}$$

where $\boldsymbol{I} \in \mathbb{R}^{m \times m}$ denotes the identity matrix. The top m principal component images of a HSI are defined as

$$\boldsymbol{I}_{\text{pci}} \in \mathbb{R}^{m \times h \times w} = \boldsymbol{W}^T \mathbb{X}. \tag{2}$$

$\boldsymbol{I}_{\text{pci}}$ are then convolved with a Gabor filter [58] with different orientations and scales. Finally, the filtering coefficients are extracted as the Gabor feature for each pixel; the Gabor feature matrix is represented as $\boldsymbol{X}_{\text{gabor}} \in \mathbb{R}^{d_2 \times n}$, where d_2 is obtained by the scales and orientations of the Gabor filter times and the number of principal component images m. In this article, the scales and orientations of the Gabor filter are set to 5 and 8, respectively. We apply Gabor filter to the top principal component image; thus, $d_2 = 5 \times 8 \times 1 = 40$.
3) *EMP Feature [52]:* The EMP feature is also based on the top m principal component images. The morphological profile for each principal component image is generated by the repeated use of openings and closings with different structural elements (SEs). The EMP feature for each pixel is then constructed by concatenating the morphological profiles; the EMP feature matrix is represented as $\boldsymbol{X}_{\text{emp}} \in \mathbb{R}^{d_3 \times n}$, where d_3 is related to the number of SEs and the number of principal component images m. In this article, we set the number of sructural elements to 4 and use the five top principal component images; thus, $d_3 = (2 \times 4 + 1) \times 5 = 45$. For the details of the EMP feature, please refer [52].
4) *EMAP Feature [53], [54]:* The EMAP feature is based on the top m principal component images and the morphological attribute filters. First, the extended attribute profiles (EAPs) for each principal component image are generated using the morphological attribute filters. The EAP is then further evolved to the EMAP with four different attributes: area of the regions, size of the regions, elongation of the regions, and homogeneity of the regions. An EAP is computed for each of these attributes. Finally, the EMAP feature for each pixel is constructed by concatenating all the attribute profiles. The EMAP feature matrix is represented as $\boldsymbol{X}_{\text{emap}} \in \mathbb{R}^{d_4 \times n}$, where d_4 is related to the parameters of the attribute filters and the number of principal component images m. As in [54], each attribute generates nine features, and we use five top principal component images; thus, $d_4 = 9 \times 4 \times 5 = 180$. For further information on the EMAP feature, please refer [53] and [54].

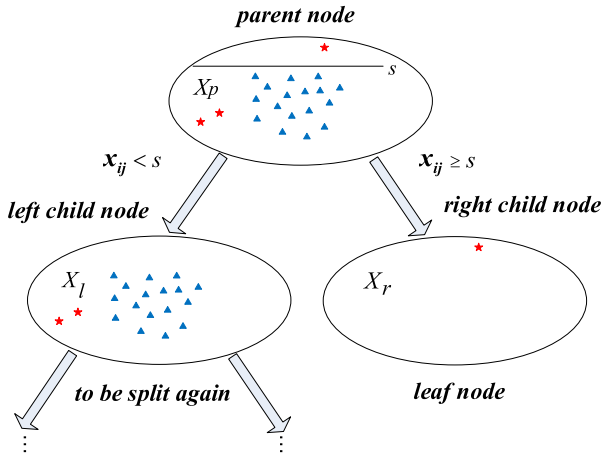


Fig. 2. One split in the isolation tree for the spectral feature.

Once these four features are extracted, we feed them into the subsequent isolation tree construction procedure.

B. Building Isolation Trees for Multiple Features of the HSI

The isolation tree was firstly proposed by Liu *et al.* [55], [56]. The structure of the isolation tree, which takes the form of a proper binary tree, is a foundational architecture of the hyperspectral anomaly detector. The proposed anomaly detector is based on the assumption that the anomalous pixels are both fewer than and different from the background pixels, meaning that they are more vulnerable to isolation. In this section, we build isolation trees for the four above-mentioned features: $\boldsymbol{X}_{\text{spe}}$, $\boldsymbol{X}_{\text{gabor}}$, $\boldsymbol{X}_{\text{emp}}$, and $\boldsymbol{X}_{\text{emap}}$.

Next, we take the spectral feature $\boldsymbol{X}_{\text{spe}}$ as an example to illustrate the construction of the isolated tree. Supposing that the sample set at the current parent node is presented in the form of the matrix $\boldsymbol{X}_p = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_p] \in \mathbb{R}^{d_1 \times p}$, where p is the number of samples in the sample set $\boldsymbol{X}_p$. To build an isolation tree, we recursively split the sample set $\boldsymbol{X}_p$ corresponding to the parent node into two parts: the sample set $\boldsymbol{X}_l \in \mathbb{R}^{d_1 \times l}$, corresponding to the left child node, and the sample set $\boldsymbol{X}_r \in \mathbb{R}^{d_1 \times r}$, corresponding to the right child node. The main issues in this process are how to partition the sample set $\boldsymbol{X}_p$ at each internal node and when to stop growing the tree.

1) Partitioning the Sample Set $\boldsymbol{X}_p$: A simple and direct method is used to partition the sample set $\boldsymbol{X}_p$ at the current parent node into two parts. First, we randomly select a band i, where $1 \leq i \leq d_1$; thus, we obtain the set $\{x_{i1}, x_{i2}, \ldots, x_{ip}\}$ from the ith row of the data matrix $\boldsymbol{X}_p$. A split value s is that the falls between the maximum and minimum values of this set are then randomly selected. Thus, the sample set $\boldsymbol{X}_p$ is splitted into $\boldsymbol{X}_l$ and $\boldsymbol{X}_r$ according to the values of the ith row

$$\begin{cases} \boldsymbol{x}_j \in \boldsymbol{X}_l, & \text{if } x_{ij} < s \\ \boldsymbol{x}_j \in \boldsymbol{X}_r, & \text{if } x_{ij} \geq s \end{cases} \tag{3}$$

where $\boldsymbol{x}_j \in \mathbb{R}^{d_1 \times 1}$ is the jth column of $\boldsymbol{X}_p$.

2) Stopping Growth: The splitting procedure is repeated recursively until either $|\boldsymbol{X}_p| = 1$ or the current isolation tree height h_c reaches the height limit h_l. h_c is defined as the

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Algorithm 1 Building an Isolation Tree for the Spectral Feature

Input: The spectral feature matrix $\boldsymbol{X}_{\text{spe}}$, the current isolation tree height h_c and the height limit h_l.
repeat
1. Randomly select a band i $(1 \le i \le d_1)$ and obtain the set $\{x_{i1}, x_{i2}, \ldots, x_{ip}\}$ for the current parent node corresponding to the sample set $\boldsymbol{X}_p$.
2. Randomly select a split value s between the maximum and minimum values of the set $\{x_{i1}, x_{i2}, \ldots, x_{ip}\}$. Thus, $\boldsymbol{X}_p$ is split into $\boldsymbol{X}_l$ and $\boldsymbol{X}_r$ according to Eq. (3).
3. If $|\boldsymbol{X}_l| > 1$ and $h_c < h_l$, $\boldsymbol{X}_l$ needs to be split again, update $\boldsymbol{X}_p$ to $\boldsymbol{X}_l$ and $h_c = h_c + 1$.
4. If $|\boldsymbol{X}_r| > 1$ and $h_c < h_l$, $\boldsymbol{X}_r$ needs to be split again, update $\boldsymbol{X}_p$ to $\boldsymbol{X}_r$ and $h_c = h_c + 1$.
until There is no sample set to be split again.
Output: An isolation tree model T_{spe}, with the band i and split value s at each internal node.

number of edges extending from the root node to the current node.

The details of this process are presented in Algorithm 1. Moreover, one split in an isolation tree for the spectral feature of hyperspectral imagery is shown in Fig. 2. The isolation trees are constructed for each of the four features such that the corresponding isolated trees t_{spe}, t_{gabor}, t_{emp}, and t_{emap} can be obtained.

C. Proposed MFIFD Algorithm

The isolation forest with multiple isolation trees for each feature is constructed based on the ensemble learning mechanism. Once this is complete, we can then use the isolation forest for each feature to build a novel MFIFD for hyperspectral anomaly detector. In this section, we describe the mechanism of the proposed MFIFD method in detail and formulate an anomaly score that is meaningful for anomaly detection.

The proposed MFIFD is a two-stage process. In the training stage, the isolation forest is built using a subsample set derived from the corresponding features: $\boldsymbol{X}_{\text{spe}}$, $\boldsymbol{X}_{\text{gabor}}$, $\boldsymbol{X}_{\text{emp}}$, and $\boldsymbol{X}_{\text{emap}}$. In the testing stage, moreover, all pixels are passed through the isolation forest for each feature to obtain an anomaly score for each pixel.

1) Training Stage: In the training stage, we build the isolation forest for the four features: $\boldsymbol{X}_{\text{spe}}$, $\boldsymbol{X}_{\text{gabor}}$, $\boldsymbol{X}_{\text{emp}}$, and $\boldsymbol{X}_{\text{emap}}$. We take the spectral feature $\boldsymbol{X}_{\text{spe}}$ as an example to illustrate the construction of an isolation forest.

We build an isolation forest with t_{spe} isolation trees for the spectral feature $\boldsymbol{X}_{\text{spe}}$, where t_{spe} denotes the number of isolation trees in the isolation forest. Each isolation tree is built by recursively splitting a subsample set $\boldsymbol{X}_\psi = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_\psi] \in \mathbb{R}^{d_1 \times \psi}$ until the pixels are isolated or until h_c reaches the height limit h_l; here, $\boldsymbol{X}_\psi$ is randomly selected from $\boldsymbol{X}_{\text{spe}} \in \mathbb{R}^{d_1 \times n}$ without replacement and $\boldsymbol{X}_\psi \subset \boldsymbol{X}_{\text{spe}}$.

There are two parameters in this stage: the subsample set size ψ and the number of isolation trees t_{spe}. ψ controls the size of the input data matrix $\boldsymbol{X}_\psi$ for each isolation tree; we empirically set $\psi = 256$ as the default value for each feature in the experiments. Moreover, t_{spe} controls the ensemble size of the isolation trees for the spectral feature; we empirically set $t_{\text{spe}} = 25$ as the default values in our experiments. This training stage is described in more detail in Algorithm 2. At the end of this stage, an isolation forest with t_{spe} isolation trees is returned ready for the testing stage.

Algorithm 2 Building an Isolation Forest for the Spectral Feature

Input: The spectral feature matrix $\boldsymbol{X}_{\text{spe}}$, the size of sub-sample set ψ and the number of isolation trees t_{spe}.
for $i = 1$ to t_{spe} **do**
1. Obtain the sub-sample set $\boldsymbol{X}_\psi$ from $\boldsymbol{X}_{\text{spe}}$.
2. Perform Algorithm 1, where the input parameters are $\boldsymbol{X}_\psi$, $h_c = 0$ and $h_l = \text{ceiling}(\log_2 \psi)$.
end for
Output: An isolation forest with t_{spe} isolation trees.

Algorithm 3 Obtaining the $h_\upsilon(\boldsymbol{x}^t_{\text{spe}})$ for the Spectral Feature

Input: The test pixel $\boldsymbol{x}^t_{\text{spe}}$, the υth isolation tree and the current isolation tree height $h_c = 0$.
repeat
1. Obtain the value $\boldsymbol{x}^t_{\text{spe}}(i)$ according to the band i at the current parent node.
2. If $\boldsymbol{x}^t_{\text{spe}}(i) < s$, the test pixel $\boldsymbol{x}^t_{\text{spe}}$ traverses to the left child node and $h_c = h_c + 1$.
3. If $\boldsymbol{x}^t_{\text{spe}}(i) \ge s$, the test pixel $\boldsymbol{x}^t_{\text{spe}}$ traverses to the right child node and $h_c = h_c + 1$.
until The test pixel $\boldsymbol{x}^t_{\text{spe}}$ traverses to a leaf node.
Output: $h_\upsilon(\boldsymbol{x}^t_{\text{spe}}) = h_c + c(|\boldsymbol{X}_\psi|) = h_c + c(\psi)$.

Finally, four isolation forests are constructed for the four features. There are five parameters in this process: ψ, t_{spe}, t_{gabor}, t_{emp}, and t_{emap}. We empirically set $\psi = 256$ and $t_{\text{spe}} = t_{\text{gabor}} = t_{\text{emp}} = t_{\text{emap}} = 25$ as the default values in our experiments.

2) Testing Stage: In this testing stage, the test pixels of the four features are represented as $\boldsymbol{x}^t_{\text{spe}} \in \mathbb{R}^{d_1 \times n}$, $\boldsymbol{x}^t_{\text{gabor}} \in \mathbb{R}^{d_2 \times n}$, $\boldsymbol{x}^t_{\text{emp}} \in \mathbb{R}^{d_3 \times n}$, and $\boldsymbol{x}^t_{\text{emap}} \in \mathbb{R}^{d_4 \times n}$, respectively. Taking $\boldsymbol{x}^t_{\text{spe}}$ as an example, the expected path length $E(h(\boldsymbol{x}^t_{\text{spe}}))$ is defined as follows:

$$E\left(h\left(\boldsymbol{x}^t_{\text{spe}}\right)\right) = \frac{1}{t_{\text{spe}}} \sum_{\upsilon=1}^{t_{\text{spe}}} h_\upsilon\left(\boldsymbol{x}^t_{\text{spe}}\right) \tag{4}$$

which is derived by passing $\boldsymbol{x}^t_{\text{spe}}$ through the t_{spe} isolation trees. The υth path length $h_\upsilon(\boldsymbol{x}^t_{\text{spe}})$ is defined as the number of edges from the root node to a terminating node when the test pixel $\boldsymbol{x}^t_{\text{spe}}$ traverses through the υth isolation tree. The process of calculating the υth path length $h_\upsilon(\boldsymbol{x}^t_{\text{spe}})$ is outlined in detail in Algorithm 3.

We can then obtain $E(h(\boldsymbol{x}^t_{\text{gabor}}))$, $E(h(\boldsymbol{x}^t_{\text{emp}}))$, and $E(h(\boldsymbol{x}^t_{\text{emap}}))$ in the same way as $E(h(\boldsymbol{x}^t_{\text{spe}}))$. The anomaly

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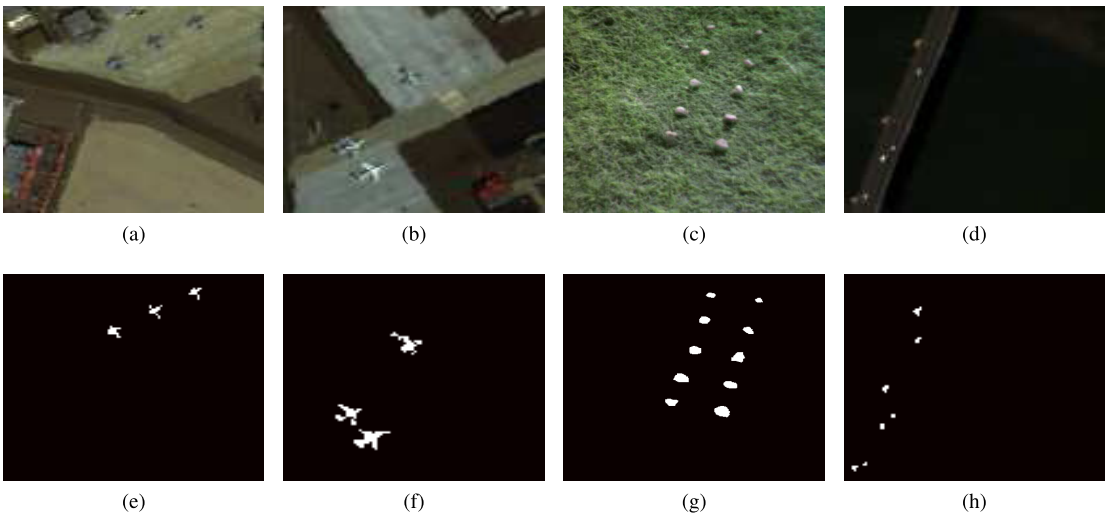


Fig. 3. Image scene descriptions. (a) False color image of the AVIRIS-I data set. (b) False color image of the AVIRIS-II data set. (c) False color image of the Cri data set. (d) False color image of the PaviaC data set. (e) Ground truth map of the AVIRIS-I data set. (f) Ground truth map of the AVIRIS-II data set. (g) Ground truth map of the Cri data set. (f) Ground truth map of the PaviaC data set.

score δ for the test pixel is defined as follows:

$$\delta = 2^{-\frac{E\left(h\left(x^t_{\text{spe}}\right)\right)+E\left(h\left(x^t_{\text{gabor}}\right)\right)+E\left(h\left(x^t_{\text{emp}}\right)\right)+E\left(h\left(x^t_{\text{emap}}\right)\right)}{4c(\psi)}} \tag{5}$$

where $c(\psi)$ denotes the average path length and is defined as follows [55], [56]:

$$c(\psi) = 2H(\psi - 1) - (2(\psi - 1)/\psi) \tag{6}$$

where $H(\psi-1)$ denotes the harmonic number and is estimated by $\ln(\psi - 1) + 0.5772156649$. To find the top m anomalous pixels, we simply sort the pixels using δ in descending order. The first m pixels are the top m anomalous pixels.

III. Experimental Results

In this section, we carry out several experiments to illustrate the detection performance of the proposed MFIFD algorithm. All experiments are implemented on a PC with E5-2680 v4 at 2.40 GHz and 256-GB RAM, MATLAB 2016b.

A. Hyperspectral Data Sets

Four different hyperspectral data sets captured from different scenes are used to evaluate the detection performance of the proposed MFIFD method. The descriptions of these data sets are as follows.

1) *AVIRIS-I Data Set:* This data set was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) from San Diego, CA, USA. The spatial resolution is 3.5 m/pixel, and the spectral resolution is 10 nm. It has 224 spectral bands in wavelengths ranging from 370 to 2510 nm. After the removal of water absorption, low signal-to-noise ratio, and poor-quality bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 bands are retained in the experiments. The whole image scene has a size of 400 × 400 pixels. A region with a size of 120 × 120 pixels is selected for testing from the top left of this image and denoted as AVIRIS-I. Three airplanes represented by 58 pixels are regarded as anomalies to be detected in the scene. The AVIRIS-I data set was obtained from [32]. Fig. 3(a) and (e) shows the false color image and the corresponding ground truth map of the AVIRIS-I data set, respectively.
2) *AVIRIS-II Data Set:* This data set was obtained from [39]. Compared with the AVIRIS-I data set, a region with a size of 100 × 100 pixels located at the center of the San Diego image scene is selected for testing and denoted as AVIRIS-II. Three airplanes represented by 134 pixels are regarded as anomalies to be detected in the scene. The false color image and the corresponding ground truth map are shown in Fig. 3(b) and (f), respectively.
3) *Cri Data Set:* This data set was acquired by the Nuance Cri hyperspectral sensor and obtained from [32]. The spectral resolution of this data set is 10 nm. The image scene has a size of 400 × 400 pixels and 46 spectral bands in wavelengths ranging from 650 to 1100 nm. Ten rocks represented by 2216 pixels are regarded as anomalies to be detected in the scene. The false color image and the corresponding ground truth map are shown in Fig. 3(c) and (g), respectively.
4) *PaviaC Data Set:* This data set was collected by the reflective optics system imaging spectrometer (ROSIS) sensor from the Pavia city center in Northern Italy. The spatial resolution is 1.3 m/pixel, and it has 102 spectral bands in wavelengths ranging from 430 to 860 nm. The whole image scene has a size of 108 × 120 pixels. The main background materials are a bridge and water. The vehicles on the bridge and the bare soil near the bridge pier, represented by 43 pixels, are regarded as anomalies to be detected in the scene. The false color image and the corresponding ground truth map are shown in Fig. 3(d) and (h), respectively.

B. Comparison Methods and Evaluation Indexes

In the experiments, the performance of the proposed MFIFD algorithm is evaluated through comparison with eight

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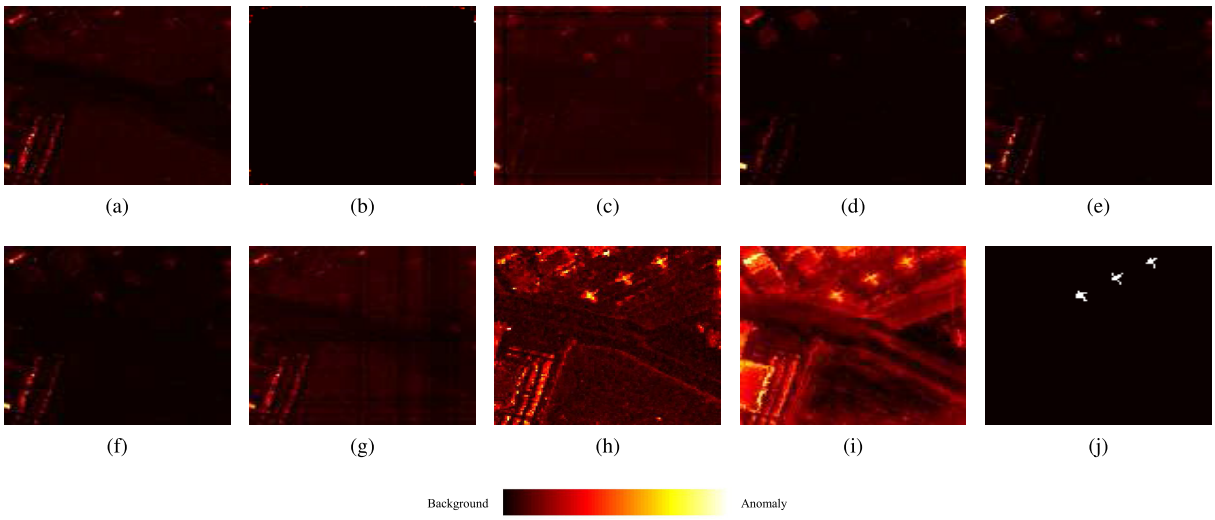


Fig. 4. Color detection maps obtained by different algorithms for the AVIRIS-I data set. (a) GRX. (b) LRX. (c) CRD. (d) LRaSMD. (e) LSMAD. (f) RPCA. (g) TenB. (h) KIFD. (i) MFIFD. (j) Ground truth map.

state-of-the-art algorithms: GRX [8], LRX [8], CRD [18], LRaSMD [31], LSMAD [32], RPCA [30], TenB [42], and KIFD [57]. Since the detection performances of the LRX and CRD algorithms are sensitive to the sliding double window sizes, the inner window size w_{in} (ranging from 3 to 23) and the outer window size w_{out} (ranging from 5 to 25) are selected optimally. For the CRD algorithm, the regularization parameter λ is set to 10^{-6}, which is consistent with the original work [18]. We further set the parameters of the LSMAD to be the same as the original work [32]: the maximal rank of the background matrix r and the cardinality of the sparse matrix k are set to 2 and 0.005, respectively. Moreover, the parameters of the LRaSMD, RPCA, TenB, and KIFD are also set to be in accordance with the original work [30], [31], [42], [57].

The anomaly detection performance of different algorithms is evaluated both qualitatively and quantitatively. The qualitative evaluation is demonstrated by the color detection map, whereas the quantitative evaluation is conducted by using the receiver operating characteristic (ROC) curve [59], the area under the ROC curve (AUC) value, the normalized background-anomaly separation map, and the running time. The ROC curve reflects the relationship between the true positive rate (TPR) and false positive rate (FPR), which are obtained by the thresholds ranging from 0 to 1. The TPR and FPR are also known as the detection probability and the false alarm rate, respectively. The TPR and FPR are defined as follows:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

$$\text{FPR} = \frac{\text{FP}}{\text{TN} + \text{FP}} \tag{8}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively. An algorithm that achieves superior performance usually has a high TPR value under the same FPR value; thus, the corresponding ROC curve will be located near the upper leftmost corner of the coordinate plane, resulting in a large area under the curve. Moreover, the AUC value refers to the AUC and is encircled by the ROC curve and the false alarm rate axis. An outstanding algorithm usually has a high AUC value.

In addition, for the normalized background-anomaly separation map, the anomaly scores for all pixels of each algorithm are normalized to within a range of 0–1 using the following formula:

$$\bar{\boldsymbol{a}}_s = \frac{\boldsymbol{a}_s - \min(\boldsymbol{a}_s)}{\max(\boldsymbol{a}_s) - \min(\boldsymbol{a}_s)} \tag{9}$$

where $\boldsymbol{a}_s \in \mathbb{R}^{n\times 1}$ denotes the anomaly scores of all pixels and $\bar{\boldsymbol{a}}_s \in \mathbb{R}^{n\times 1}$ denotes the normalized anomaly scores. $\max(\boldsymbol{a}_s)$ and $\min(\boldsymbol{a}_s)$ represent the minimum and maximum values of $\boldsymbol{a}_s$, respectively. Once this process is complete, the normalized anomaly score distributions of the background and anomalous pixels are depicted using a box plot. The plotted results of a good algorithm typically feature a distinct gap between the background box and the anomaly box. Meanwhile, the background normalized anomaly scores are suppressed to be within a small range. The running time is also considered to facilitate an evaluation of computational complexity of different algorithms.

C. Detection Performance

For the AVIRIS-I data set, the color detection maps of different algorithms are shown in Fig. 4. The proposed MFIFD and KIFD algorithms can accurately detect the shapes and locations of the three airplanes although some false anomalies are also detected. The CRD, LSMAD, and RPCA algorithms can detect the locations of the three airplanes, but the shapes of the airplanes are not rendered accurately. The GRX, LRX, LRaSMD, and TenB algorithms obtain a poor separation between the anomalous and background pixels and cannot detect the anomalous pixels effectively. Moreover, the ROC curves and the corresponding AUC values are shown in Fig. 5. From the figure, it can be seen that the curves of the proposed MFIFD and KIFD algorithms are closer to the top-left corner

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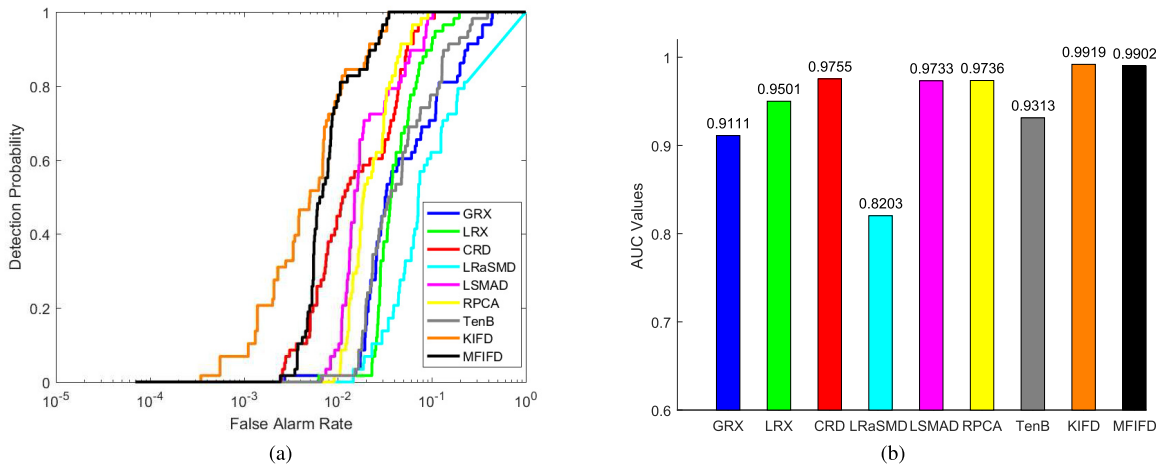


Fig. 5. Detection accuracy evaluation for the AVIRIS-I data set. (a) ROC curves. (b) AUC values.

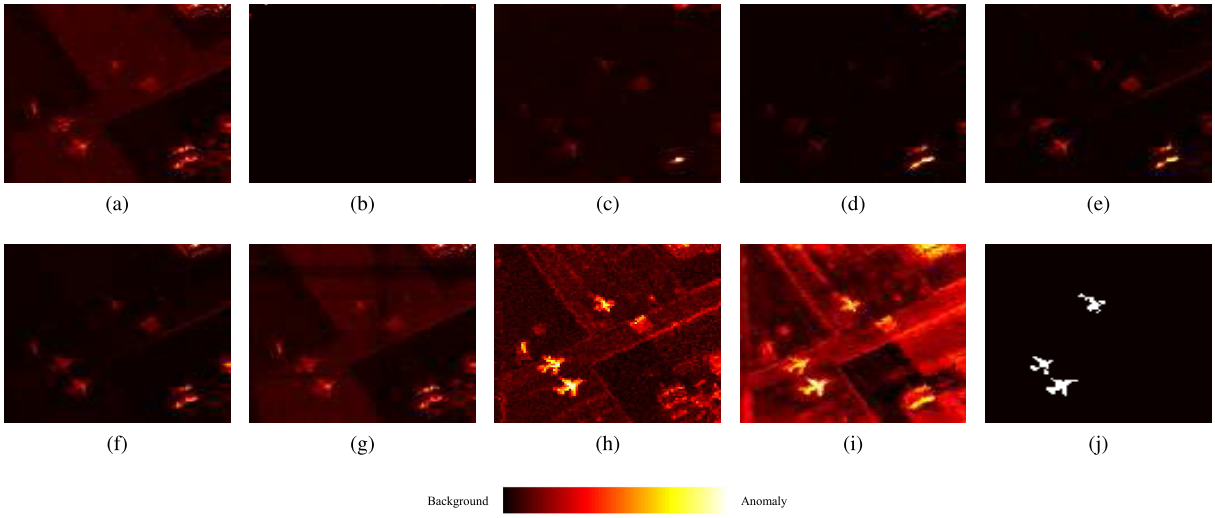


Fig. 6. Color detection maps obtained by different algorithms for the AVIRIS-II data set. (a) GRX. (b) LRX. (c) CRD. (d) LRaSMD. (e) LSMAD. (f) RPCA. (g) TenB. (h) KIFD. (i) MFIFD. (j) Ground truth map.

than the others; in addition, the AUC values of the proposed MFIFD and KIFD algorithms are 0.9902 and 0.9919, which are larger than the other algorithms.

For the AVIRIS-II data set, the color detection maps of different algorithms are shown in Fig. 6. The LRX, CRD, and LRaSMD algorithms fail to detect the anomalies in this scene and mistakenly identify some background pixels as anomalies. The GRX, LSMAD, RPCA, and TenB algorithms can detect the locations of the three airplanes, but the shapes of the airplanes are blurred. For its part, the MFIFD and KIFD algorithms can detect the locations and shapes of the three airplanes from among the background pixels to a greater degree than the GRX, LSMAD, RPCA, and TenB algorithms. Fig. 7 shows the ROC curves of the different algorithms and their corresponding AUC values. We can observe from the figure that the ROC curves of the MFIFD and KIFD algorithms are closer to the top-left corner than the others; moreover, the AUC values of the MFIFD and KIFD algorithms are 0.9837 and 0.9916, which are larger than the other algorithms.

For the Cri data set, the color detection maps of different algorithms are shown in Fig. 8. The CRD, LSMAD, KIFD, and MFIFD algorithms can effectively detect the locations and shapes of the ten rocks, whereas the GRX, LRX, RPCA, and TenB algorithms can detect their locations but blur some of their shapes. The LRaSMD algorithm cannot detect anomalies effectively in this scene. Fig. 9(a) further shows that the CRD, LSMAD, KIFD, and MFIFD algorithms all obtain high detection probabilities with a low false-alarm rate, whereas Fig. 9(b) shows that the proposed MFIFD algorithm obtains a higher AUC value than the other algorithms.

For the PaviaC data set, the color detection maps of different algorithms are shown in Fig. 10. The proposed MFIFD algorithm can accurately detect the locations and shapes of the anomalous pixels although some background pixels are mistakenly identified as anomalous pixels. The GRX, LRX, CRD, LSMAD, RPCA, and TenB algorithms can only detect the locations of some anomalous pixels, and the shapes of some anomalous pixels are missing. The LRaSMD and KIFD algorithms cannot detect anomalies effectively in this scene. In Fig. 11(a), the ROC curves demonstrate that the proposed MFIFD algorithm obtains a higher detection probability over the other detectors. The AUC values of the different algorithms

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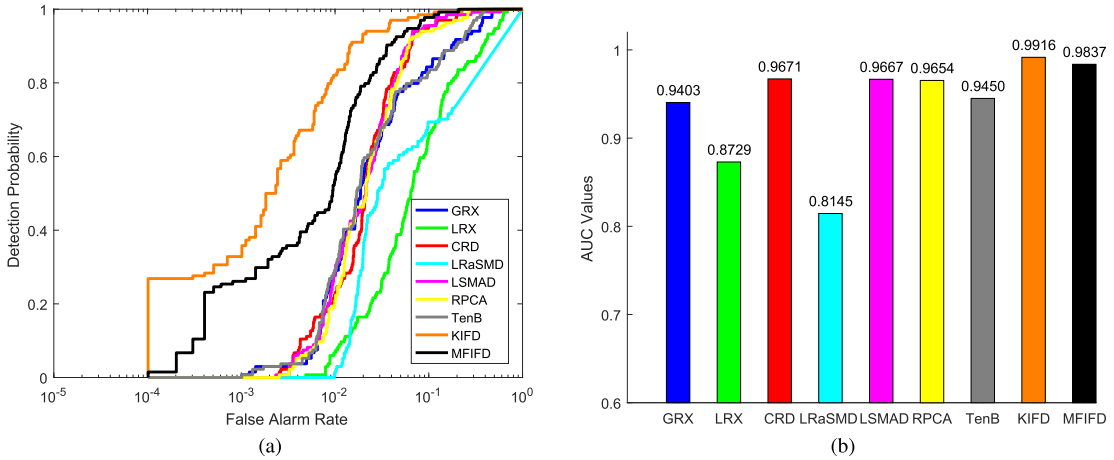


Fig. 7. Detection accuracy evaluation for the AVIRIS-II data set. (a) ROC curves. (b) AUC values.

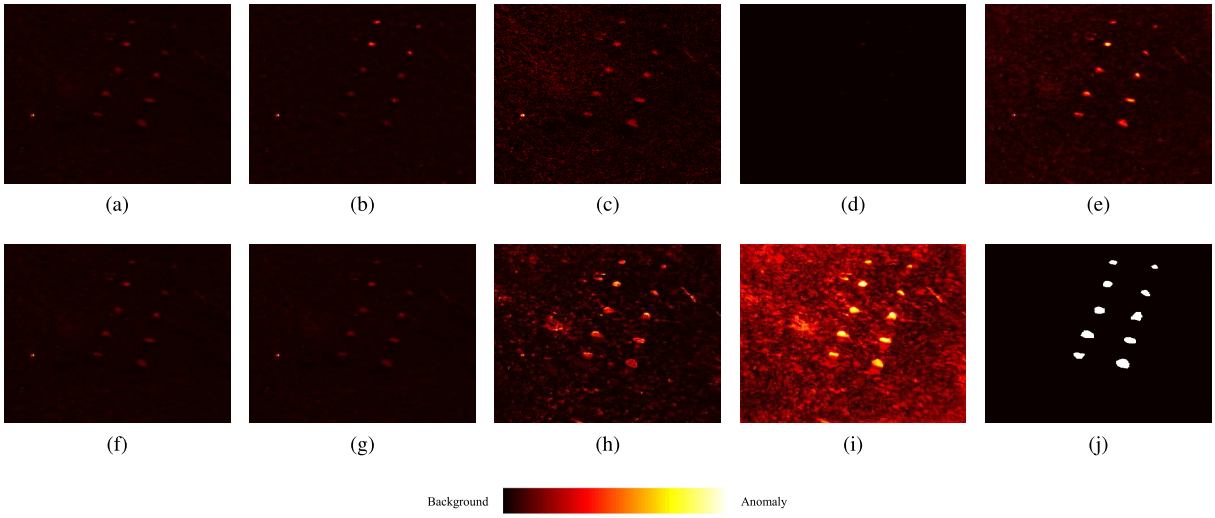


Fig. 8. Color detection maps obtained by different algorithms for the Cri data set. (a) GRX. (b) LRX. (c) CRD. (d) LRaSMD. (e) LSMAD. (f) RPCA. (g) TenB. (h) KIFD. (i) MFIFD. (j) Ground truth map.

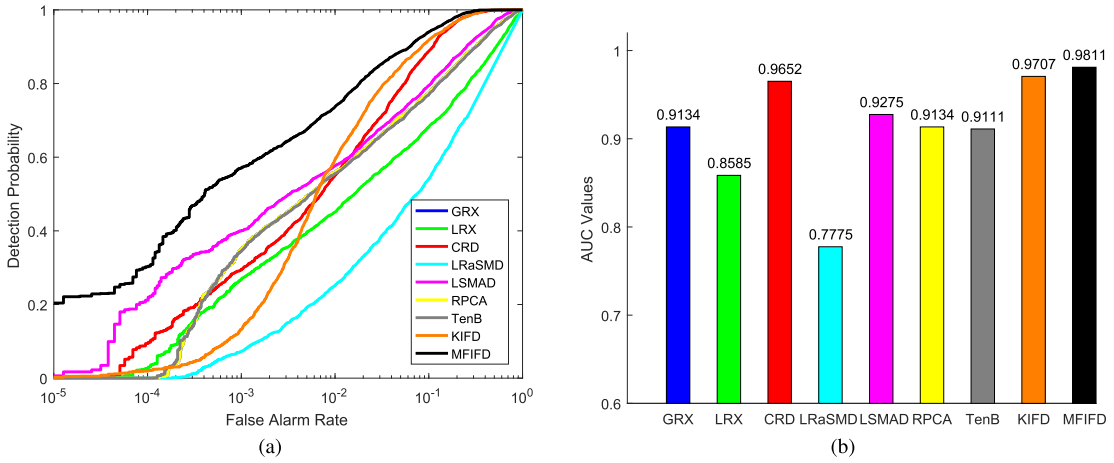


Fig. 9. Detection accuracy evaluation for the Cri data set. (a) ROC curves. (b) AUC values.

are shown in Fig. 11(b); these values indicate that the proposed MFIFD algorithm can achieve the best detection results out of all compared algorithms.

To facilitate a more in-depth evaluation of the proposed MFIFD algorithm's detection performance, the normalized background-anomaly separation maps for the four data sets are shown in Fig. 12. There are two boxes for each algorithm: the cyan box represents the distribution of the background, whereas the red box indicates the distribution of the anomaly. The position of these two boxes reflects the algorithm's

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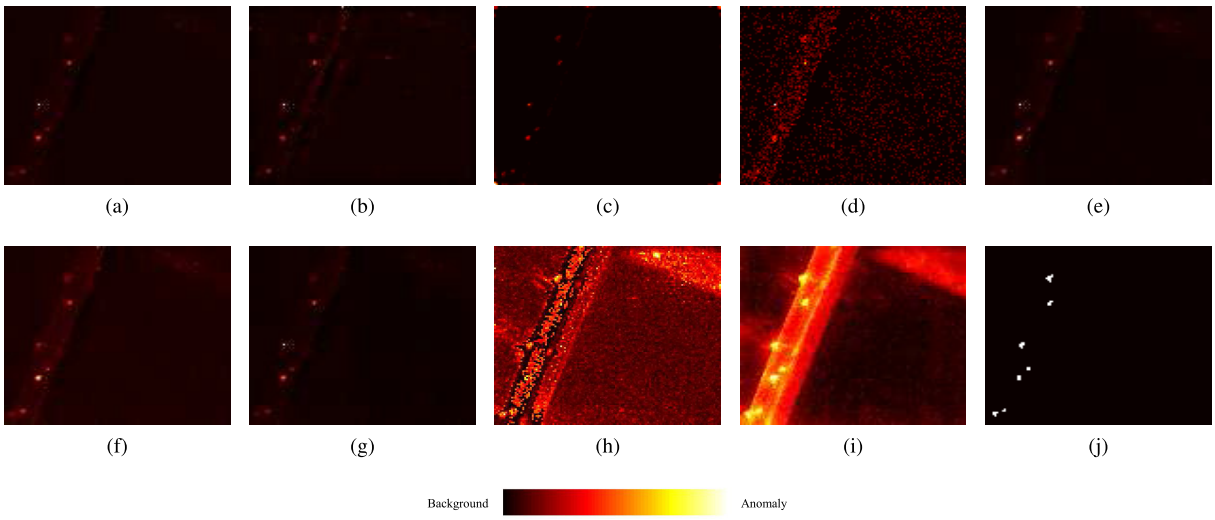


Fig. 10. Color detection maps obtained by different algorithms for the PaviaC data set. (a) GRX. (b) LRX. (c) CRD. (d) LRaSMD. (e) LSMAD. (f) RPCA. (g) TenB. (h) KIFD. (i) MFIFD. (j) Ground truth map.

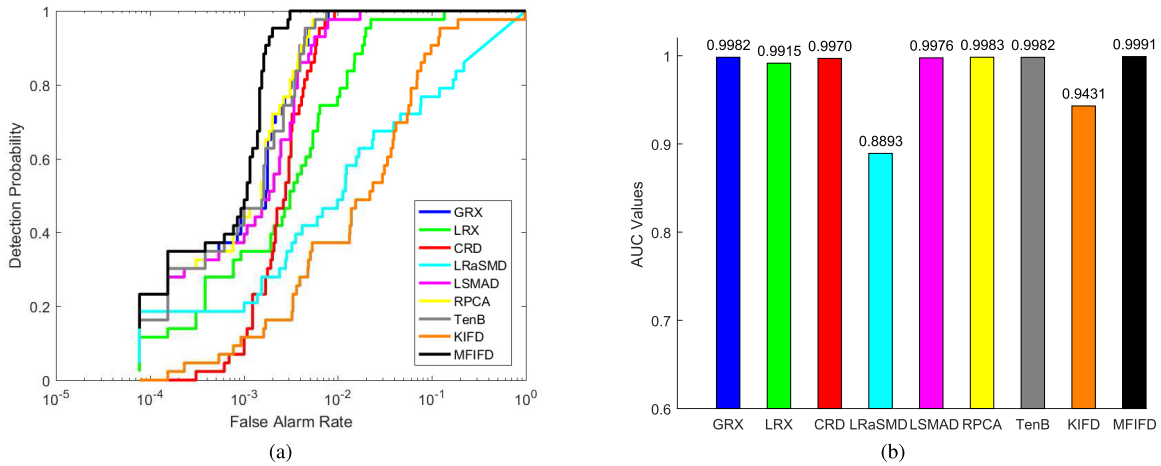


Fig. 11. Detection accuracy evaluation for the PaviaC data set. (a) ROC curves. (b) AUC values.

ability to distinguish between background and anomaly pixels. In other words, a larger gap between the two boxes indicates that the corresponding algorithm achieves a superior detection performance.

Fig. 12(a) shows the separation maps for the AVIRIS-I data set. We can see that the separation gaps for the proposed MFIFD and KIFD algorithm are larger than those for the other algorithms; meanwhile, the background is also validly compressed. This indicates that the MFIFD and KIFD algorithms achieve a better separation ability. In addition, the CRD, LSMAD, and RPCA algorithms also exhibit good separation ability although the results of the GRX, LRX, LRaSMD, and TenB algorithms are comparatively worse. The separation maps for the AVIRIS-II data set are shown in Fig. 12(b). Here, the proposed MFIFD and KIFD algorithms still achieve larger separation gaps than the other algorithms. Moreover, the separation abilities of the GRX, CRD, LSMAD, RPCA, and TenB algorithms are slightly poorer since their separation gaps are narrower than that of the MFIFD and KIFD algorithms. In contrast with the aforementioned algorithms, the LRX and LRaSMD algorithms perform relatively unsatisfactorily. For the Cri data set, it can be seen from Fig. 12(c) that overlapping has occurred in almost all algorithms. Among them, the CRD and LRaSMD algorithms achieve similar separation results, whereas the GRX, LRX, LSMAD, RPCA, TenB, and KIFD algorithms obtain relatively better separation results. The proposed MFIFD algorithm demonstrates the best separation ability. Finally, Fig. 12(d) shows the separation maps for the PaviaC data set. Almost all algorithms show distinct separation gaps. Among them, the LRX, LRaSMD, and KIFD algorithms have similar separation results, whereas the GRX, CRD, LSMAD, RPCA, TenB, and MFIFD algorithms obtain relatively better separation results.

In addition, the AUC values of all algorithms for the four data sets are shown in Table I, whereas the running times of all algorithms for the four data sets are shown in Table II. As shown in Table II, the GRX algorithm is very fast, with a running time much lower than that of the other algorithms. It is worth noting that the running time of the proposed MFIFD algorithm is similar to that of the GRX algorithm; meanwhile,

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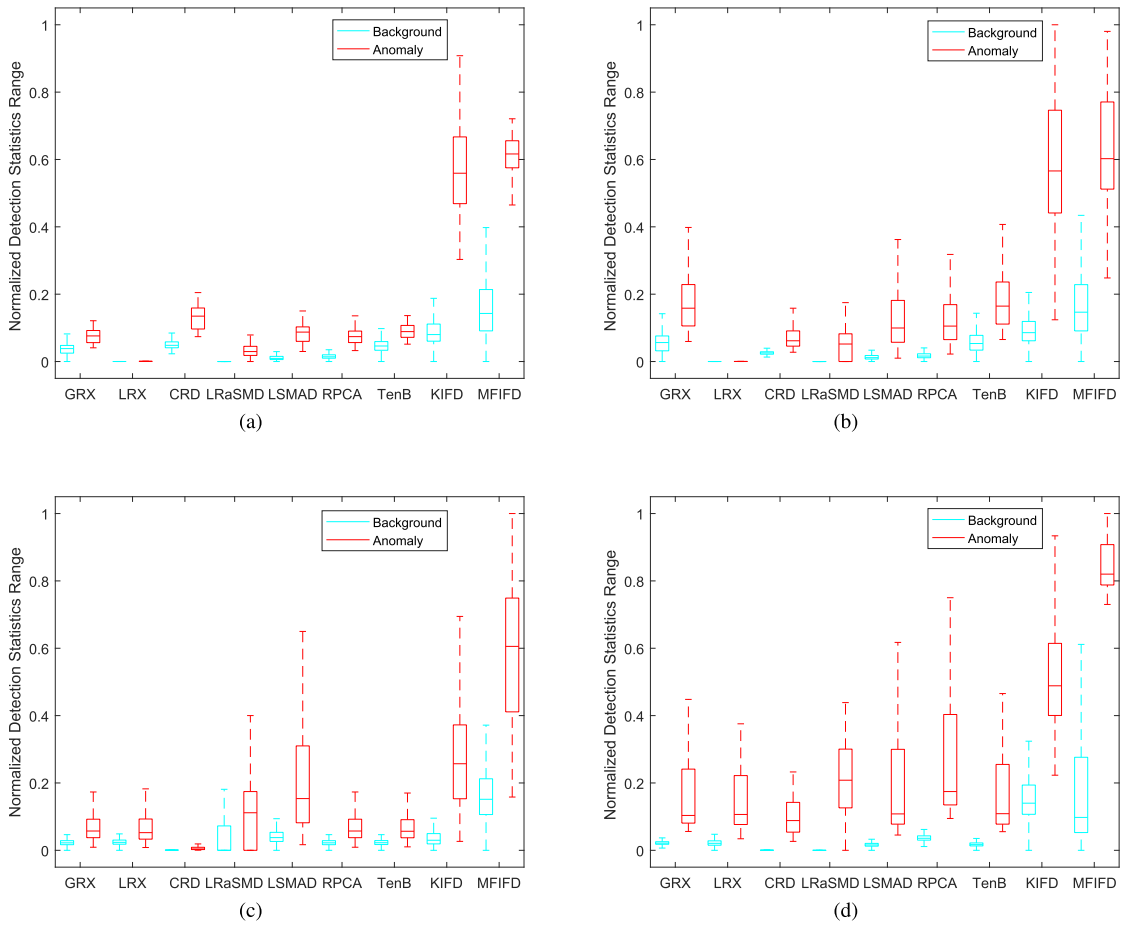


Fig. 12. Normalized background-anomaly separation maps of different algorithms for the four data sets. (a) AVIRIS-I data set. (b) AVIRIS-II data set. (c) Cri data set. (d) PaviaC data set.

TABLE I
AUC VALUES FOR ALL ALGORITHMS

Dataset	GRX	LRX (w_{in}, w_{out})	CRD (w_{in}, w_{out})	LRaSMD	LSMAD	RPCA	TenB	KIFD	MFIFD
AVIRIS-I	0.9111	0.9501 (13, 25)	0.9755 (19, 23)	0.8203	0.9733	0.9736	0.9313	0.9919	0.9902
AVIRIS-II	0.9403	0.8729 (13, 25)	0.9671 (19, 21)	0.8145	0.9667	0.9654	0.9450	0.9916	0.9837
Cri	0.9134	0.8585 (21, 25)	0.9652 (13, 17)	0.7775	0.9275	0.9134	0.9111	0.9707	0.9811
PaviaC	0.9982	0.9915 (9, 11)	0.9970 (9, 17)	0.8893	0.9976	0.9983	0.9982	0.9431	0.9991

TABLE II
RUNNING TIME (SECONDS) FOR ALL ALGORITHMS

Dataset	GRX	LRX (w_{in}, w_{out})	CRD (w_{in}, w_{out})	LRaSMD	LSMAD	RPCA	TenB	KIFD	MFIFD
AVIRIS-I	0.32	128.35 (13, 25)	93.54 (19, 23)	22.20	23.01	28.65	1.65	76.45	2.41
AVIRIS-II	0.17	96.44 (13, 25)	25.56 (19, 21)	14.77	15.58	18.32	1.08	66.92	1.79
Cri	0.74	180.12 (21, 25)	551.95 (13, 17)	55.44	51.18	68.15	4.43	436.63	7.75
PaviaC	0.12	22.15 (9, 11)	130.35 (9, 17)	9.72	9.67	11.26	0.79	62.05	1.75

the proposed MFIFD algorithm also achieves outstanding detection performance. Furthermore, the LRaSMD, LSMAD, and RPCA algorithms are more time-consuming than GRX, TenB, and MFIFD algorithms; in addition, LRX, CRD, and KIFD consume more running time than the other algorithms.

D. Parameter Analysis and Discussion

In this section, the parameter analysis of the proposed MFIFD algorithm is undertaken on the four data sets. There are two parameters in the proposed MFIFD algorithm: namely, the size of subsample set ψ and the number of isolation trees for the four features (t_{spe}, t_{gabor}, t_{emp}, and t_{emap}). In this article, the number of isolation trees for each feature is set to the same value: $t_{\text{spe}} = t_{\text{gabor}} = t_{\text{emp}} = t_{\text{emap}}$. Here, the AUC value and running time is adopted to assess the proposed MFIFD algorithm.

Fig. 13 shows the effect of different subsample set sizes on the detection performance and running time for each data

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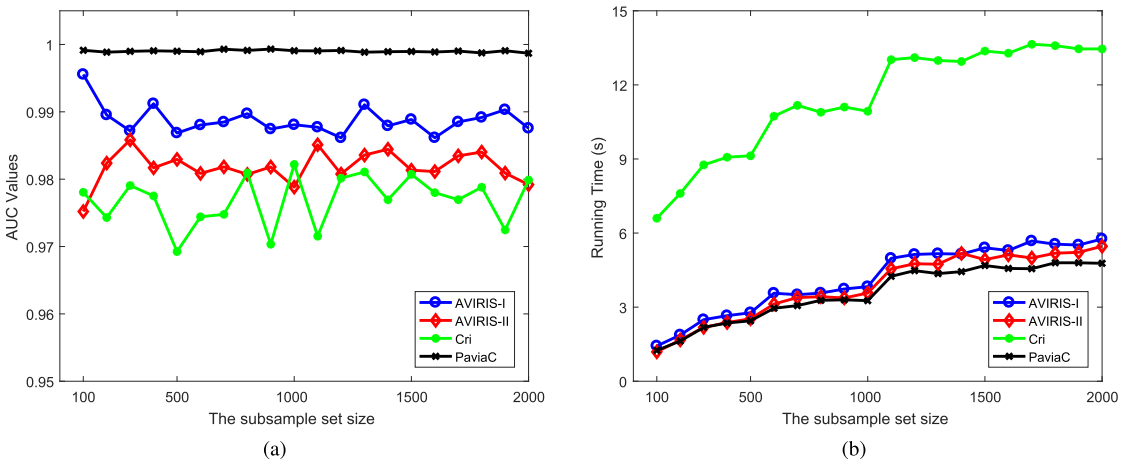


Fig. 13. Effect of the subsample set size on each data set. (a) AUC values. (b) Running time.

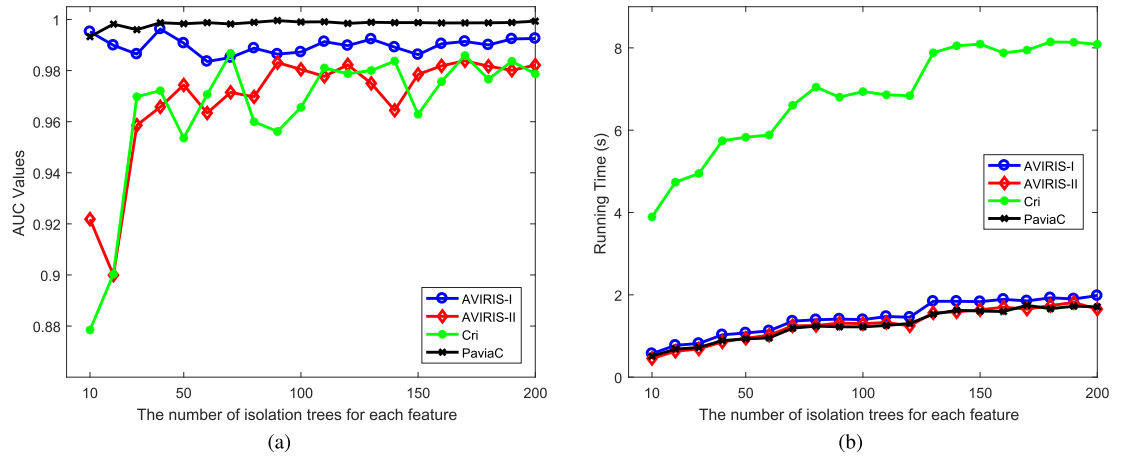


Fig. 14. Effect of the number of isolation trees for each feature on each data set. (a) AUC values. (b) Running time.

set. The value of the subsample set size is varied in the range $[100, 200, \ldots, 2000]$. As shown in Fig. 13(a), the AUC values for the PaviaC data set are nearly stable, whereas the AUC values for the AVIRIS-I, AVIRIS-II, and Cri data sets fluctuate slightly within a small range from 0.97 to 0.995. In Fig. 13(b), we can observe that the running time for each data set achieves nearly linear growth when the subsample set size increases. In the next step, the effect of different numbers of isolation trees for each feature on the four data sets is analyzed. The value of the number of isolation trees for each feature is changed within the range of $[10, 20, \ldots, 200]$. In Fig. 14(a), we see that the AUC values for the PaviaC data set are nearly stable, whereas the AUC values for the AVIRIS-I data set fluctuate slightly within a small range from 0.98 to 1.00. In addition, the AUC values for the AVIRIS-II and Cri data sets increase at first and then fluctuate within a small range. As shown in Fig. 14(b), we can observe that the running time for each data set achieves nearly linear growth when the number of isolation trees for each feature increases. Therefore, by considering both the performance and efficiency of the MFIFD algorithm, we set $\psi = 256$ and $t_{\text{spe}} = t_{\text{gabor}} = t_{\text{emp}} = t_{\text{emap}} = 25$ as the default parameters in this article.

IV. Conclusion

In this article, a novel multiple feature and isolation forest-based detector (MFIFD) for the HAD task is proposed. The spectral, Gabor, EMP, and EMAP features are used to build the corresponding isolated forest. The advantages of the proposed MFIFD algorithm are twofold: first, it can make the full use of the spectral and spatial information of the HSI by the combination of multiple spectral and spatial features; second, it is a fast hyperspectral anomaly detector suitable for the real-world applications. Experiments on four real hyperspectral data sets demonstrate that MFIFD can outperform other state-of-the-art algorithms in terms of both qualitative and quantitative evaluations. Moreover, the parameter sensitivity analysis proves the robustness of the proposed MFIFD algorithm under reasonable parameter settings.

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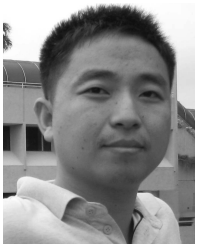
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